Branch: Computer Science and Engineering **Class:** B.Tech. 5th Sem.
Course Title: Operating Systems **Course Code:** PCC CS-502 **Credits:** 3 **L-T-P:** 3-0-0
Total Marks: 100 **Internal Marks:** Theory-50 Practical-0 **External Marks:** 100

Course Objectives:

- Introduction to the fundamental understanding of operating system principles and Functionality.
- Introductions to the key concepts of an operating system, such as processes, scheduling, Synchronization, Deadlocks, Memory management, and I/O management.
- Introduction to the important design and development considerations for an operating system.
- Introduction to additional operating system topics and features.
- Introduction to applying operating system concepts in software development and system administration.
- Introduction to problem-solving skills in the context of operating system design and implementation.

Course Outcomes:

- Understand the structure and organization of file systems and OS design.
- Understand processes and how they are synchronized and scheduled.
- Analyze various memory management techniques.
- Understand system calls for managing processes, memory & file systems, and the algorithms used in OS implementation.

Unit I:

Introduction to operating systems-operating system functions, Evolution of operating systems, Batch, Interactive, Time sharing and Real-time system, System protection, Operating system structure, Operating system services, System calls and System programs.

Process management: Process concept, Process state diagram, Schedulers, Scheduling concept, Scheduling criteria, Scheduling algorithms, Multiprocessor scheduling.

Unit II:

Concurrent processes-principle of concurrency, Producer-consumer problem, Critical section problem, Semaphores, Inter-process communication, Classical problems in concurrency.

Deadlock: System model, Deadlock characterization, Prevention, Avoidance, Detection and Recovery.

Unit III:

Memory management-resident monitor, Contiguous memory management techniques, Non-contiguous allocation - Paging, Segmentation, Performance. Allocation strategies.

Virtual memory management: Virtual memory concept, Demand paging, Performance, Page replacement algorithms, Allocation of frames, Thrashing.

Unit IV:

I/O Management & Disk Scheduling: I/O devices and organization of I/O function, I/O buffering, DISK scheduling, and operating system design issues.

File System: File system overview, File organization and Access mechanism, File directories, File sharing.

Books Recommended:

Text Books:

1. Operating System Concepts by A. Silberschatz, P. Galvin, And G. Gagne, John Wiley and Sons.
2. Design of the UNIX Operating System By M. J. Bach, Prentice Hall of India, 1986.
3. Stallings, William. OperatingSystems: Internals and Design Principles. Upper Saddle River, Nj: Prentice Hall, 2001.

References:

1. Operating System Concepts by J. Peterson, A. Silberschatz, And P. Galvin, Addison Wesley. Operating Systems by Milan Milenkovic, Tata McGraw Hill.

E-Resources: